

JGR Solid Earth

RESEARCH ARTICLE

10.1029/2023JB026648

Key Points:

- A new graph-based measure that identifies dead-end fractures is proposed
- Late-time solute tailing is maximized at the percolation threshold
- The density of dead-end fractures is a good predictor of non-Fickian transport

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Citation:

Yoon, S., Hyman, J. D., Han, W. S., & Kang, P. K. (2023). Effects of dead-end fractures on non-Fickian transport in three-dimensional discrete fracture networks. *Journal of Geophysical Research: Solid Earth*, 128, e2023JB026648. <https://doi.org/10.1029/2023JB026648>

Received 2 MAR 2023

Accepted 6 JUL 2023

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Effects of Dead-End Fractures on Non-Fickian Transport in Three-Dimensional Discrete Fracture Networks

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Abstract Understanding mechanistic causes of non-Fickian transport in fractured media is important for many hydrogeologic processes and subsurface applications. This study elucidates the effects of dead-end fractures on non-Fickian transport in three-dimensional (3D) fracture networks. Although dead-end fractures have been identified as low-velocity regions that could delay solute transport, the direct relation between dead-end fractures and non-Fickian transport has been elusive. We systematically generate a large number of 3D discrete fracture networks with different fracture length distributions and fracture densities. We then identify dead-end fractures using a novel graph-based method. The effect of dead-end fractures on solute residence time maximizes at the critical fracture density of the percolation threshold, leading to strong late-time tailing. As fracture density increases beyond the percolation threshold, the network connectivity increases, and dead-end fractures diminish. Consequently, the increase in network connectivity leads to a reduction in the degree of late-time tailing. We also show that dead-end fractures can inform about main transport paths, such as the mean tortuosity of particle trajectories. This study advances our mechanistic understanding of solute transport in 3D fracture networks.

Plain Language Summary A critical phenomenon often observed in fractured rocks is an anomalously slow decay of solute concentration, which is often called solute tailing. Long retention of solute contaminants poses severe threats to human health, and predicting solute tailing is critical for remediation purposes. However, solute tailing is challenging to predict, and understanding the mechanisms inducing solute tailing is important for improving our predictive capability. Recent studies showed that dead-end fractures form slow-flow regions that could cause solute tailing. However, no study has explicitly quantified the effects of dead-end fractures on solute tailing. By simulating flow and transport in three-dimensional fracture networks over more than 200 realizations, we find that the relationship between fracture density and dead-end fracture formation is non-monotonic. Specifically, we observe that there exists a critical level of fracture density where the presence of dead-end fractures is maximized. At the optimum level, solute particles' residence time in dead-end fractures significantly increases, leading to strong solute tailing. As fracture density increases beyond the optimum, dead-end fractures diminish due to the improved network connectivity, reducing solute tailing. This study improves our mechanistic understanding of how dead-end fractures control solute tailing in fracture networks.

1. Introduction

Understanding transport processes in fractured rocks is critical for various subsurface applications and hydrogeological processes such as contaminant transport, in-situ carbon mineralization, nuclear waste repository design, and geothermal engineering (Abdelghani et al., 2015; Berkowitz, 2002; Follin et al., 2014; Parker et al., 2012; Pérez-Flores et al., 2017; Snæbjörnsdóttir et al., 2020; Tsang et al., 2015; Watanabe & Takahashi, 1995). A Fickian framework can effectively model solute transport in relatively homogeneous systems. However, fracture networks inherently have high structural complexity, such as dead-end fractures and multiple clusters of percolating fractures. The inherent structural complexity leads to complex flow and transport processes, making non-Fickian transport the norm rather than an anomaly in fracture networks (Becker & Shapiro, 2000; Berkowitz & Scher, 1997; Edery et al., 2016; Geiger et al., 2010; Hyman, Rajaram, et al., 2019; Kang, Le Borgne, et al., 2015; Moreno et al., 1990; Roubinet et al., 2013; Sherman et al., 2021).

Writing – review & editing: Seonkyoo Yoon, Jeffrey D. Hyman, Weon Shik Han, Peter K. Kang

The structural complexity of fracture networks makes it challenging to identify the detailed mechanisms that lead to non-Fickian transport. Recent advances in computing power and the development of discrete fracture network (DFN) models are now opening new possibilities for detailed investigation of flow and transport processes in three-dimensional (3D) fractured media with higher levels of fidelity (de Dreuzy et al., 2012; Hyman et al., 2021; Kang et al., 2020; Maillot et al., 2016). In particular, the influence of geostructural attributes such as fracture density and fracture length distribution on non-Fickian transport can be explicitly investigated using DFN models.

Recent studies used 3D DFNs to investigate how fracture density controls flow channelization (Hyman, 2020; Hyman, Dentz, et al., 2019; Sherman et al., 2019, 2020). Flow channelization is a phenomenon where fluid flow occurs primarily through a few transmissive and well-connected fractures and is known to result in anomalously early arrival of tracers. Hyman (2020) found that low fracture density leads to significant flow channelization, which controls solute transport by constraining the solute particle trajectories to a few preferential flow channels. Further, a wide range of solute transport behavior is observed in sparse networks, including a power-law tailing of solute arrival times, a key feature of non-Fickian transport. As fracture density increases, network connectivity increases and flow channelization is shown to diminish. The improved connectivity homogenizes flow distribution, which in turn leads to a faster decay rate of solute tailing (Hyman, 2020; Sherman et al., 2020).

In addition to flow channeling, low-velocity regions within dead-end fractures have been suggested as a potential source of non-Fickian transport. While the dead-end fractures are no-flow regions in two-dimensional fracture networks, there can be fluid flow through dead-end fractures in 3D networks. Flow velocities in 3D dead-end fractures are likely slow (Hyman et al., 2017; Hyman & Jiménez-Martínez, 2018; Kang et al., 2020; Park et al., 2003). Thus, the dead-end fractures have been expected to impact late-time transport behavior (Hyman et al., 2017; Kang et al., 2020; Srinivasan et al., 2018). Hyman et al. (2017) showed that a subnetwork model that disregards minor fractures not participating in the main transport path could capture the mean behavior of solute transport. However, the subnetwork model requires calibration for predicting late-time transport behavior, implying that minor fractures, such as dead-end fractures, may control late-time transport behavior. Although the previous studies have shown that dead-end fractures can be a potential source of late-time tailing, there has been no systematic study that explicitly quantifies the effects of dead-end fractures on late-time solute tailing. In particular, the influence of dead-end fractures on solute transport at different fracture densities has been unclear.

This study aims to establish the mechanistic understanding of dead-end fracture effects on solute transport, especially in late-time regimes. Primarily, we address the following two major questions: (a) How does the fracture density control the formation of dead-end fractures? (b) What is the role of dead-end fractures on non-Fickian transport behaviors such as solute tailing? The paper is structured as follows. In Section 2, we explain the methods for fracture network generation, dead-end fracture identification, and the simulation of flow and transport. In Section 3, we present and discuss the effects of fracture density on dead-end fracture formation and how the dead-end fracture impacts transport. We provide conclusions in Section 4.

2. Methodology

We simulate flow and solute particle transport in fracture networks that are systematically generated over wide ranges of fracture density and fracture length distributions. We employ a novel topology-based method, as detailed in Section 2.2, to identify dead-end fractures. We simulate flow and transport in the generated network realizations using *dfnWorks*, which is a high-fidelity modeling suite for 3D DFN modeling (Hyman, Karra, et al., 2015). This methodology section consists of three subsections describing network generation, graph-based dead-end fracture identification, and numerical simulation of fluid flow and solute transport. The script files for generating a DFN, identifying dead-end fractures, and running flow and particle transport simulations have been made publicly available (Yoon et al., 2023).

2.1. Generating Fracture Networks

We generate 3D DFNs with various fracture densities and fracture length distributions in a cubic domain of length $L = 15$ m. Recently, progresses have been made in generating fracture networks via kinematic and geomechanical models that can simulate fracture nucleation and growth (Davy et al., 2013; Lavoine et al., 2020; Paluszny et al., 2020; Thomas et al., 2020; Viswanathan et al., 2022). However, the use of geomechanically-generated fractures is beyond the scope of this study because creating a large ensemble of fractures with many fractures is

not yet pragmatic with the new methodologies. We generate an ensemble of networks using a stochastic Poisson process where circular-shaped fractures are randomly placed into the network with random orientations. We assigned a buffer region outside the model domain where fractures can be generated. When an outside fracture (generated in the buffer region) is connected to a percolating fracture cluster, the fracture is truncated by the domain's boundaries and kept in the fracture network. Thus, the local fracture intensity near the boundaries is comparable with internal ones. The resultant networks mimic disordered fracture networks observed in the field (Klint et al., 2004).

Fractures are represented as polygons with eight vertices considering planar discs whose radii r follow a truncated power-law distribution with decay exponent α . The power-law distributed fracture length is commonly observed in the field (Bonnet et al., 2001). The truncated power-law distribution is written as:

$$p_r(r, r_0, r_u) = \frac{\alpha}{r_0} \frac{(r/r_0)^{-1-\alpha}}{1 - (r_u/r_0)^{-\alpha}}. \quad (1)$$

The lower cutoff, r_0 , is set to 1 m and the upper cutoff, r_u , is set to 5 m. The decay exponent α determines the tail thickness of the length distribution. The probability density in the tail region controls the frequency of large fractures; larger fractures are likely to be sampled with lower exponents. We consider three different values of $\alpha = [2.0; 2.2; 2.4]$ that are observable in the field (Bonnet et al., 2001).

We define the hydraulic fracture aperture b by correlating it to the fracture radius r using a positively correlated power-law relationship:

$$b = \gamma r^\beta, \quad (2)$$

where $\gamma = 5.0 \times 10^{-4}$ and $\beta = 0.5$. The unit of the factor γ is $m^{0.5}$. The power-law relation between fracture length and aperture is often assumed in DFN models and based on field observations (Bogdanov et al., 2007; de Dreuzy et al., 2002; Frampton & Cvetkovic, 2010; Hyman et al., 2016; Joyce et al., 2014; Wellman et al., 2009). Note that hydraulic apertures are uniform within fractures, but vary between them. While previous studies have established that variations in aperture roughness within a fracture lead to flow channeling (Boutt et al., 2006; Cardenas et al., 2007; Detwiler et al., 2000; Kang et al., 2016), the importance of roughness on network-scale channelization has yet to be fully understood. Although it is possible to include aperture variations into three-dimensional DFNs (de Dreuzy et al., 2012; Frampton et al., 2019; Hyman et al., 2021), we do not include them in these simulations for several reasons. First, while it is known that aperture variations can cause local flow channelization, there is also evidence that there are circumstances where internal fracture aperture variability has little effect on upscaled flow and transport properties, namely when the roughness is low variance and has short correlation lengths (Makedonska et al., 2016). Additionally, to include aperture variability into high-fidelity simulations requires detailed knowledge of the rock formation, the fracture formation process, and any hydraulic and geochemical alteration, for example, dissolution and precipitation. Recall that our networks are meant to be semi-generic and cannot be appropriately conditioned to such properties. With this in mind, the results and conclusions reported here are restricted to physical scenarios where in-fracture variability effects ought to be negligible relative to the controls imposed by the network structure (Hyman, 2020).

Fracture density varies widely in nature (Davy et al., 2013) and governs the overall connectivity of fracture networks. To investigate the effects of fracture density on the formation of dead-end fractures and solute transport, we consider a range of normalized density which is defined as $\rho = \frac{p}{p_c}$ where p is the number of fractures in a DFN, and the normalizing factor, p_c , is a critical percolation threshold. The percolation threshold is the minimum number of fractures required for a fracture cluster to span or percolate the whole domain (Berkowitz & Balberg, 1993; Bour & Davy, 1997, 1998; Sahimi, 1994). Using the percolation threshold as the normalizing factor helps measure fracture density with respect to the percolation threshold regardless of fracture length distribution and the shape of fractures. Due to the advantages, the dimensionless definition of fracture density has been used in previous studies (Hyman, 2020; Hyman, Dentz, et al., 2019; Sherman et al., 2020).

In an infinite medium, the percolation probability $\prod_L(p)$ can be expressed using the Heaviside function $\mathcal{H}$ as $\prod_L(p) = \mathcal{H}(p - p_c)$, meaning that the percolation occurs as a sharp transition. However, in finite systems with a domain size L , the percolation probability $\prod_L(p)$ shows a smooth transition from 0 to 1 as p increases. The smooth transition can be described by the following error function (Thovet et al., 2017),

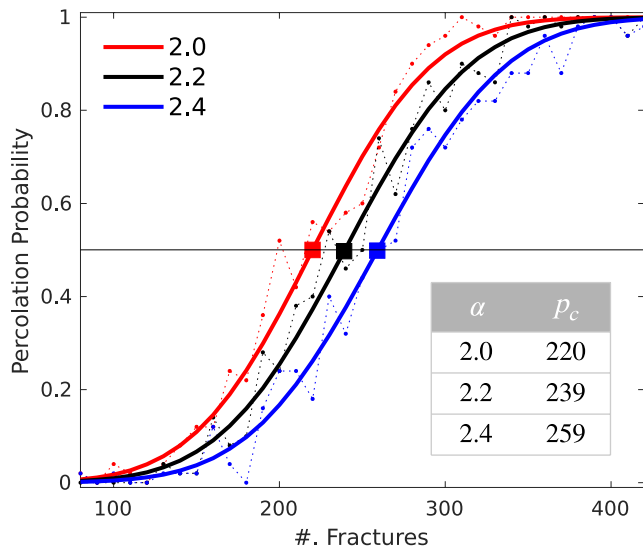


Figure 1. Percolation probability as a function of p . The dotted lines are the empirical percolation probability. The solid lines are the error function in Equation 3 fitted to the dotted lines. The inset table shows the estimated percolation thresholds according to α .

$$\prod_L(p) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^p \exp\left\{-\frac{(\varphi - p_c)^2}{2\sigma^2}\right\} d\varphi, \quad (3)$$

where σ is the scale parameter related to the range of p within which the probability transition occurs, and the critical percolation threshold p_c is defined such that $\prod_L(p_c) = 1/2$ (Thovert et al., 2017). For each α , we generate 50 network realizations for a range from 10 to 500 in increments of 10 to estimate percolation thresholds. We compute the empirical percolation probability for each case of p , and fit the error function in Equation 3 to the empirical function, estimating the threshold p_c for each α (Thovert et al., 2017). The rigorous identification of percolation thresholds allows us to generate DFNs systematically with different densities.

Figure 1 shows the fitted function (Equation 3) of percolation probabilities for different α , and the inset table presents the estimated thresholds. We consider five different fracture densities: $\rho = [0.5; 1; 2; 4; 8]$. For all combinations of α and ρ , we generate $N_e = 15$ realizations of discrete fracture networks (DFNs) to avoid case-specific results. This results in a total of $225 \left(3_{(\#,\alpha)} \times 5_{(\#,\rho)} \times 15_{(N_e)} \right)$ fracture networks. In a DFN realization, isolated fractures that are not connected to inlet or outlet planes may exist. We disregard the isolated fractures in all analyses because they do not participate in flow and transport. Note that fracture density ρ is determined by dividing the total number of fractures by the critical percolation threshold, irrespective

of whether the fractures are connected or not. Figure 2 shows examples of fracture networks, excluding isolated fractures, for the combinations of $\alpha = [2.0; 2.2; 2.4]$ and $\rho = [0.5; 1; 4]$.

2.2. Graph Representation and Dead-End Fracture Identification

Graph theory provides a powerful framework for analyzing the topology of fracture networks and identifying dead-end fractures (Aldrich et al., 2017). We first describe how a graph represents a fracture network, and propose a novel way of identifying dead-end fractures using a graph-based method.

In a graph representation, a fracture is represented by a vertex, and an intersection between two fractures is represented by an edge (Hyman et al., 2017). When the fracture network F consists of n_F fractures as $F = \{f_i\}$ for $i \in \mathbf{i}$ where $\mathbf{i} = \{1, \dots, n_F\}$, we can derive a mapping function, Φ , that converts F into a graph $G(V, E)$ comprising $n = |V(G)|$ vertices and $m = |E(G)|$ edges, where $V(G)$ and $E(G)$ denote the sets of nodes and edges in G , respectively. For every $f_i \in F$, a unique vertex $u_i \in V$ exists,

$$\Phi : f_i \rightarrow u_i. \quad (4)$$

If there is an intersection between two fractures f_i and f_j , then an edge in E exists and connects the corresponding vertices u_i and u_j as

$$\Phi : f_i \cap f_j \neq \emptyset \rightarrow e_{ij} = (u_i, u_j), \quad (5)$$

where $\emptyset$ denotes empty set, and $(u_i, u_j) \in E$ denotes the edge connecting the vertices u_i and u_j . Source s and target t vertices are included into G to incorporate flow direction. We convert inlet and outlet boundaries into the source and target vertices, where each fracture intersecting the inlet boundary x_0 is connected to s ,

$$\Phi : f_i \cap x_0 \neq \emptyset \rightarrow e_{si} = (s, u_i), \quad (6)$$

and each fracture intersecting the outlet boundary x_L is connected to t ,

$$\Phi : f_i \cap x_L \neq \emptyset \rightarrow e_{it} = (u_i, t). \quad (7)$$

Figure 3 shows the graph representation of the fracture networks shown in the first row of Figure 2, where $\rho = [0.5; 1; 4]$ and $\alpha = 2.0$.

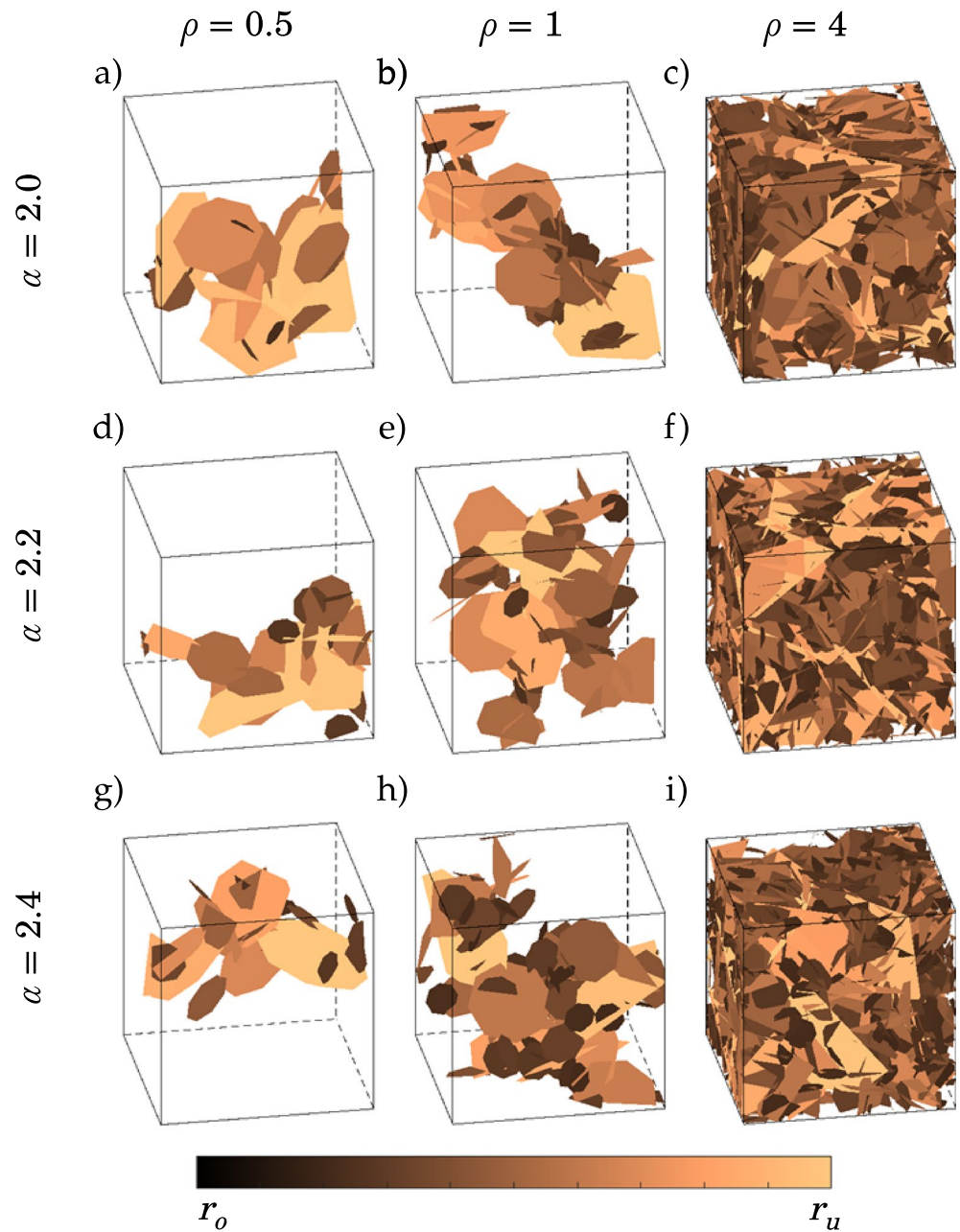


Figure 2. Examples of fracture networks for $\alpha = [2.0; 2.2; 2.4]$ and $\rho = [0.5; 1; 4]$. Each column is for a fixed value of ρ , and each row is for a fixed value of α . Fractures are color-coded such that smaller fractures are darker.

The graph representation of fracture networks allows for identifying dead-end fractures based on their topology. In this study, dead-end fractures are defined as fractures whose corresponding nodes in a graph G form a subset of the biconnected components. Removing these fractures does not disconnect the source and target vertices. A biconnected component refers to a maximal subgraph G' that does not contain any cut vertex dividing it. In simpler terms, a biconnected component is a subgraph G' that remains connected even if any vertex $u \in G$ is removed. The proposed definition of dead-end fractures corresponds to a cluster of fractures where solute can only enter from and exit through a single fracture, which acts as a cut vertex in the graph G .

Table 1 provides an algorithm to identify the dead-end fractures in a fracture network F . We transform F to a graph G and decompose G into biconnected components using a sequential method known as linear-time depth-first search (Tarjan & Vishkin, 1985). We denote the biconnected components whose removal does not

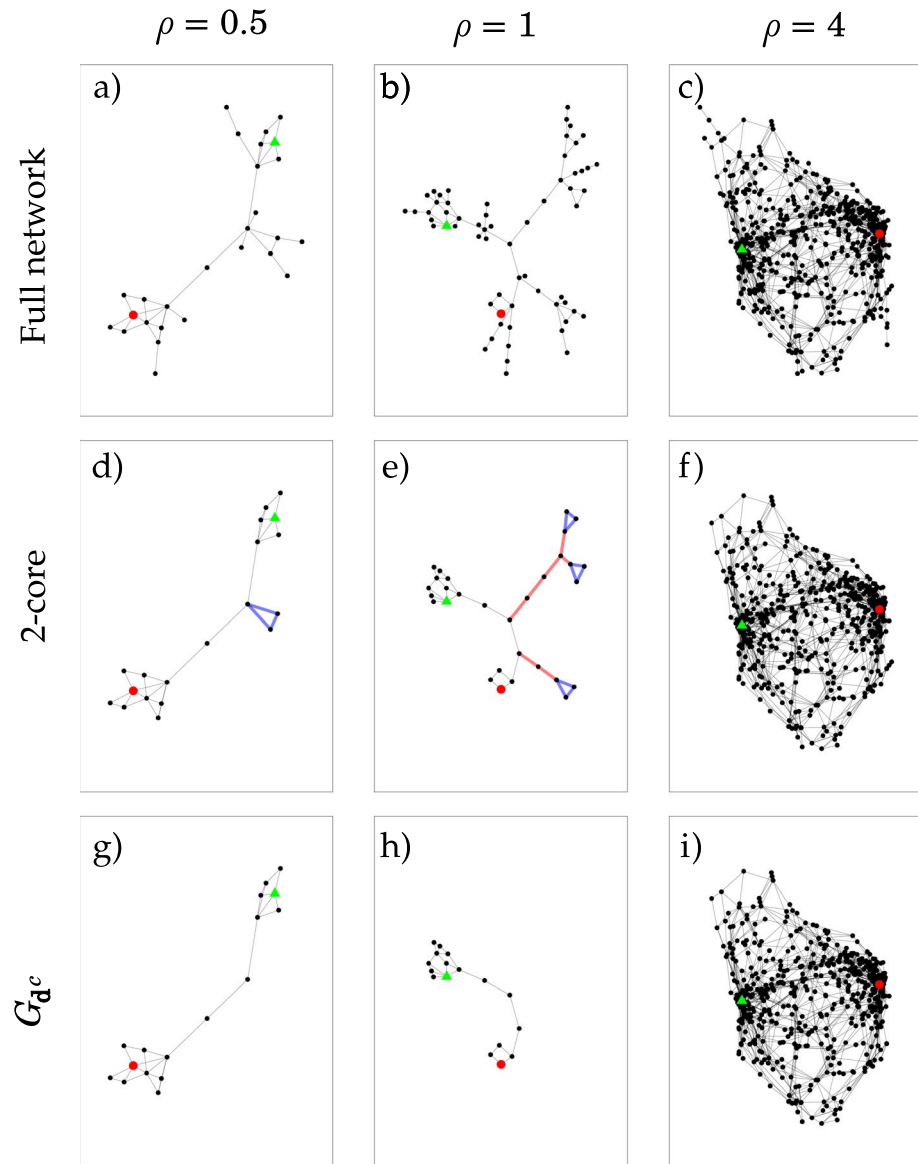


Figure 3. Graph representation of the networks shown in the first row of Figure 2 where $\rho = 0.5$ (first column), $\rho = 1$ (second column), and $\rho = 4$ (third column). Three types of graph representations are presented: full network graph (first row), 2-core subgraph of the full network graph G_{2c} (second row), and the subgraph excluding dead-end fractures G_{dc} (third row). The colored (blue or red) edges in 2-core subgraph is where solute should enter from and exit to a single node. The green triangle and red circle are the source and target nodes, respectively.

disconnect the source and target vertices as G_b for $b \in \mathbf{b}$ (see Table 1 for computing $\mathbf{b}$). The fractures whose corresponding nodes are the subset of $\mathbf{d} = \bigcup_b V(G_b)$ are the dead-end fractures. The subgraph, $G_{dc} = G - \mathbf{d}$, simplifies the full graph G and identifies the main body of the network, which should be the backbone of fracture networks for flow and transport. Here, the superscript c denotes set complement.

The complement of the 2-core subgraph, denoted as G_{2c} , has been previously proposed and widely used as a graph-based definition of dead-end fractures. In order to highlight the distinctive properties of the proposed definition of dead-end fractures, we compare it with the 2-core subgraph-based definition. The k -core of a graph is defined as the maximal subgraph containing the vertices of degree equal to or greater than k (Seidman, 1983). The 2-core subgraph-based approach defines dead-end fractures as fractures whose corresponding nodes are not part of the 2-core subgraph. Physically, the nodes in the 2-core subgraph represent fractures where solutes can

Table 1
Graph-Based Dead-End Fracture Identification

Input: a fracture network F of n_F fractures as $F = \{f_i\}$ for $i \in i$ where $i = \{1, \dots, n_F\}$

Output: dead-end fractures $F_d = \{f_d\}$ where $d \in \mathbf{d}$ and $\mathbf{d}$ is the set of nodes corresponding to dead-end fractures.

1. $\mathbf{b} = \emptyset$
2. Transform F into G by the mapping Φ in Equations 4–7
3. Decompose G into biconnected components G_m such that $G = \bigcup_m G_m$.
4. Loop for $\forall m$
 - $\mathbf{b} \leftarrow \mathbf{b} \cup m$ if $\{s, t\} \in V(G - E(G_m))$
5. $\mathbf{d} = \bigcup_b V(G_b)$ for $b \in \mathbf{b}$

enter and exit through different intersections (Doolaege et al., 2020; Hyman et al., 2017). The identification of the 2-core subgraph involves recursively removing all vertices with a degree less than two.

Figure 3e shows the 2-core subgraph G_{2c} of the full network graph shown in Figure 3b. The 2-core subgraph includes the nodes (red- and blue-colored) that can be considered as branches or minor fractures to the primary paths between source and target nodes. These branches are retained in the 2-core subgraph because they are attached to a biconnected component (blue-colored) with a minimal degree of two. In contrast, the newly proposed definition of dead-end fractures excludes these branches, as shown in Figure 3h. Thus, the proposed definition of dead-end fracture is more comprehensive and effectively captures the minor fractures that do not form part of the primary flow and transport pathways, which will be further demonstrated in the following sections. Previous studies often identify dead-end fractures using flow information, which can be computationally demanding to obtain (Doolaege et al., 2020; Maillot et al., 2016). The flow-based approach identifies low-flow fractures as dead-end fractures (Maillot et al., 2016), which implies that flow criteria (threshold) must be defined to differentiate between low-flow and high-flow regimes. The determination of these criteria is subjective and can vary depending on the specific realization of the fracture network. On the other hand, the newly proposed definition only requires the topology information of fracture networks, enabling the efficient and consistent identification of dead-end fractures.

We identify the dead-end fractures in all generated fracture networks using the proposed method in Table 1 and investigate how the extended definition can capture the dead-end fracture effects on solute transport. For this, we need transport information through the dead-end fractures, for which the following section presents the method for fluid flow and particle transport simulation.

2.3. Flow and Solute Transport Simulation

Flow in fractured rocks are often assumed laminar, and in such case, the Stokes equations can be used to model fluid flow (Bear, 1988). Since fracture size is typically significantly larger than an aperture, the Stokes equation can be reduced to the Reynolds equation (Zimmerman & Bodvarsson, 1996),

$$\nabla \cdot (b^3 \nabla P) = 0. \quad (8)$$

In this study, we assumed an impervious matrix domain, thereby neglecting matrix flow. While this assumption is generally valid for hard rocks such as granite, it is important to note that fracture-matrix interactions may play an important role on transport processes. We apply a pressure difference boundary condition of 1 MPa in the direction of the x -axis, and the other sides are assumed no-flow boundaries. We numerically solve Equation 4 with the boundary conditions using the finite volume solver implemented in PFLOTRAN (Hammond et al., 2014), determining pressure fields over the whole domain. From the pressure field, the Eulerian velocity field $\mathbf{u}(\mathbf{x})$ is reconstructed within the fracture networks (Makedonska et al., 2015; Painter et al., 2012).

We simulate solute particle transport through the network by tracing particle pathlines from the velocity field $\mathbf{u}(\mathbf{x})$. In these simulations, we exclusively consider advective transport and do not account for molecular diffusion or retention caused by matrix diffusion or adsorption. This study focuses on advective process alone rather than

simulating all possible relevant physical processes in solute transport, which complicates the interpretation. In due course and subsequent studies, additional physical processes such as diffusion and sorption and then their integrated effects will be characterized.

We inject solute particles at the inlet boundary, and the initial particle positions, denoted as $\mathbf{a}$, are allocated by flux-weighting, which means that the number of particles at a position in the inlet boundary is proportional to the inflow flux occurring at that position (Hyman, Painter, et al., 2015; Kang et al., 2017; Kreft & Zuber, 1978). We inject 10^5 particles in each simulation. We confirmed that the results were not affected by injecting more particles beyond this value.

The particle trajectory $\mathbf{x}(t; \mathbf{a})$ of a particle is given by solving the advection equation

$$\frac{d\mathbf{x}(t; \mathbf{a})}{dt} = \mathbf{v}_t(t; \mathbf{a}), \mathbf{x}(0; \mathbf{a}) = \mathbf{a}, \quad (9)$$

where $\mathbf{x}(0; \mathbf{a}) = \mathbf{a}$ indicates that the particle starts at $\mathbf{a}$ at time $t = 0$, and the Lagrangian velocity $\mathbf{v}_t(t; \mathbf{a})$ is determined as $\mathbf{v}_t(t; \mathbf{a}) = \mathbf{u}[\mathbf{x}(t; \mathbf{a})]$. At fracture intersections, particles choose an outgoing fracture with a probability proportional to the volumetric flux of the outgoing fracture (Berkowitz et al., 1994; Kang, Le Borgne, et al., 2015; Park, Lee, & Berkowitz, 2001; Park et al., 2003; Sherman et al., 2019; Stockman et al., 1997).

3. Results and Discussions

3.1. Transport Dependence on Fracture Density

We simulate flow and particle transport in all realizations of 3D DFNs generated as described in Section 2.1. Then, we record the first passage times, t_L , of particles at the outlet plane ($x = L$). The arrival times are nondimensionalized as $t'_L = Qt_L/V$, and referred to as pore volume (PV), where Q is the total volumetric influx at the inlet plane, and V is the void volume of an entire fracture network (Cardenas et al., 2007; Wang & Cardenas, 2014; Yoon & Kang, 2021). For each combination of α and ρ , we compute the PDF of t'_L , which we term first passage time distribution (FPTD) and denote it as p_{FPT} . The FPTD is obtained by averaging over the individual particle arrival times as

$$p_{\text{FPT}}(t) = \langle \delta[t - t'_L] \rangle, \quad (10)$$

where the angular brackets denote the average over all injected particles. Here, the total number of particles is $n_p = 15 \times 10^5$ because we have 15 DFN realizations for each combination of α and ρ , and we injected 100,000 particles for each simulation.

Figure 4 shows p_{FPT} for all fracture density and fracture length distribution combinations. To quantify the late-time solute tailing, we estimate the tail slopes of the FPTDs by linear regression, as shown in Figure 4. We fit a linear regression model to the range between 95% and 99.99% quantiles of the FPTDs (Yoon & Kang, 2023). Figure 5a shows the slope as a function of ρ for $\alpha = [2.0; 2.2; 2.4]$. Interestingly, the solute tailing is maximized at the percolation threshold ($\rho = 1$). As the fracture density ρ increases beyond the critical density, the degree of solute tailing decreases and transitions to Fickian transport. This result indicates the presence of critical fracture density that maximizes late-time tailing. Notably, non-Fickian transport theory identifies FPTDs with tail slopes smaller than three as non-Fickian (Dentz et al., 2004), which is the case for fracture density $\rho \leq 2$ in this study. As the fracture density ρ goes beyond two, the tail slopes drop significantly, falling in Fickian regimes (Figure 5a).

Strong late-time tailing implies that the arrival times at the outlet plane are broadly distributed, and a large probability density exists at the tail regions of the FPTDs (Aquino et al., 2015). Fracture density effects on solute spreading can be further confirmed by computing the second moments of the FPTDs defined as $(n_p - 1)^{-1} \sum (t'_L - \langle t'_L \rangle)^2$, as shown in Figure 5b. Note that the second-moment analysis is useful for highlighting the heterogeneity of particle travel times, a key characteristic of anomalous transport. The variance of the FPTDs shows a similar trend with the tail slope: it decreases dramatically as the fracture density ρ goes beyond two. Moreover, the reduced variance means that higher fracture density limits the variability of the arrival times, indicating that the complexity of network structural properties that control solute transport is likely decreasing as fracture density increases. These results demonstrate the significant effects of fracture density on solute transport. The next sections delve into how dead-end fractures play an important role in the relation between fracture density and transport.

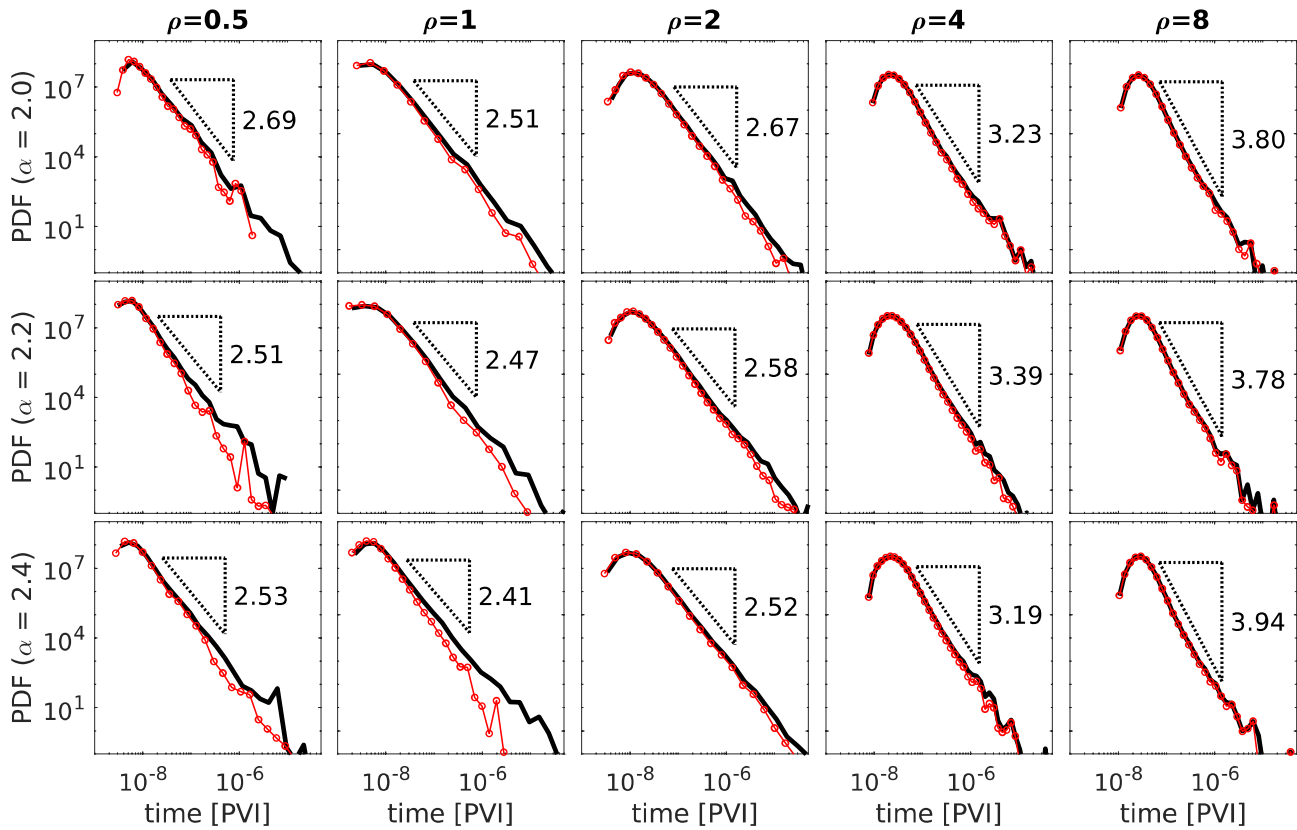


Figure 4. First passage time distributions (black-colored) and backbone residence time distributions (red-colored) for all combinations of $\rho = [0.5; 1; 2; 4; 8]$ and $\alpha = [2.0; 2.2; 2.4]$. The tail slopes are estimated by fitting a linear regression model to 95% and 99.99% quantiles.

3.2. Effects of Fracture Density on Dead-End Fractures

To investigate how fracture density determines the existence of dead-end fractures, we estimate the total surface area of dead-end fractures, $S_d = \sum_d S_d$, where S_d is the surface area of the dead-end fracture f_d such that $d \in d$ or equivalently $u_d \notin V(G_{d^c})$. We then compute the normalized surface area of the dead-end fractures' surface

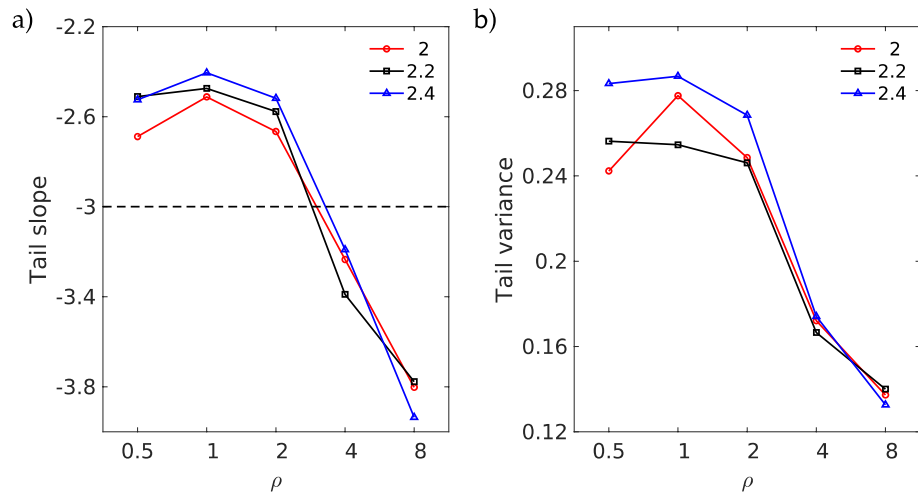


Figure 5. (a) Tail slope and (b) variance of FPTDs as a function of ρ for $\alpha = [2.0; 2.2; 2.4]$.

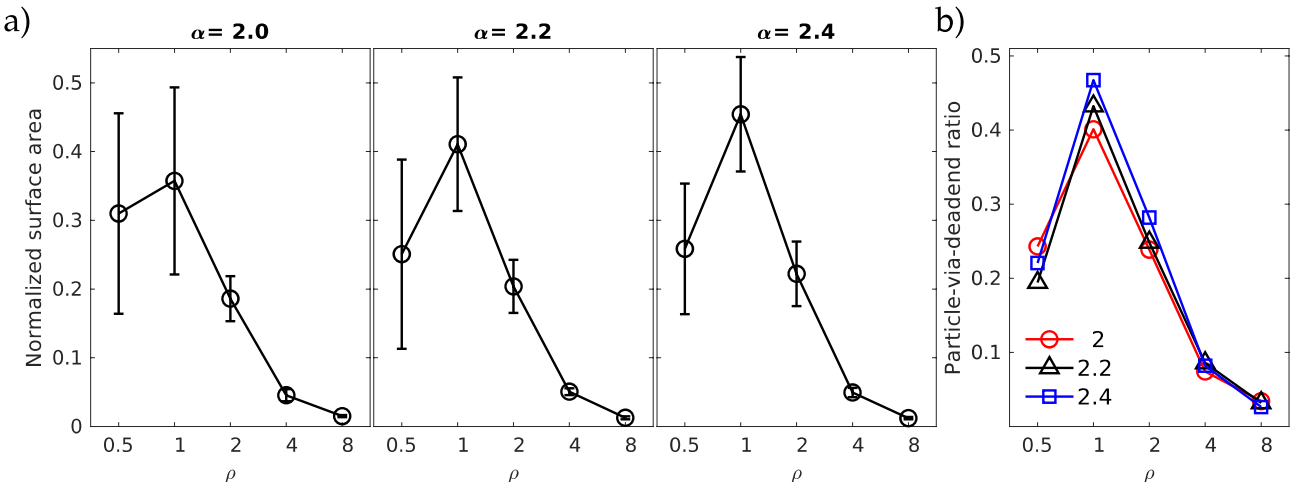


Figure 6. (a) Mean and standard deviation of the normalized area of dead-end fractures, $\bar{S}_d$. The ensemble size for each combination of α and ρ is 15. (b) The percentage of solute particles that experience dead-end fractures.

areas as $\bar{S}_d = S_d/S_i$, where $S_i = \sum_i S_i$ is the total area of all fractures and $i = \{1, \dots, n_F\}$. Figure 6a shows $\bar{S}_d$ according to fracture density. $\bar{S}_d$ increases as the fracture density increases up to the critical percolation threshold. Then the normalized surface area decreases as the fracture density goes beyond the threshold. The $\bar{S}_d$ being maximum at the percolation threshold aligns well with the solute-tailing behavior (Figure 5). We also quantified the number of dead-end fractures as a function of fracture density. When normalizing the number of dead-end fractures with the total number of fractures, the normalized number of dead-end fractures showed similar trends as the trend of dead-end fractures' surface area (not shown).

The existence of the optimum at the percolation threshold can be physically explained. When the fracture density is lower than the percolation threshold (e.g., $\rho = 0.5$), fractures must be distributed efficiently to induce percolation. If a very sparse network percolates, we can expect that most of the fractures in the network would take part in the main transport pathway. The most efficient case is that each fracture is connected only to another single fracture, and the one-to-one connection continues from the inflow to the outflow boundaries. In such an efficient

case, the fracture network has no dead-end fractures because all fractures comprise the main transport path. As the fracture density increases, fractures do not necessarily have to be most efficiently distributed to percolate because there is enough number of fractures. Thus, one can expect the number of dead-end fractures to increase. However, as the fracture density further increases beyond the critical density, the fracture connectivity will correspondingly increase. In this situation, the dead-end fractures will diminish due to the improved connectivity, leading to the presence of optimum fracture density for the number of dead-end fractures, as supported by Figure 6a.

Figure 6a also shows that the standard deviation of $\bar{S}_d$ decreases as fracture density increases. This implies that the network variability is larger in sparse networks, which leads to a wide range of flow and transport behavior, including non-Fickian transport. On the contrary, the improved connectivity in higher fracture density limits the variability of $\bar{S}_d$, which will lead to the reduced variability in flow and transport properties between realizations (Hyman, 2020; Sherman et al., 2020).

From Figures 5 and 6, we can observe a strong correlation between the formation of dead-end fractures and late-time tailing. To further elucidate the relation between dead-end fractures and solute tailing, we present the scatter plot of tail slopes and normalized dead-end fracture's area, as shown in Figure 7. The figure clearly illustrates that the tail slope rises as the normalized surface

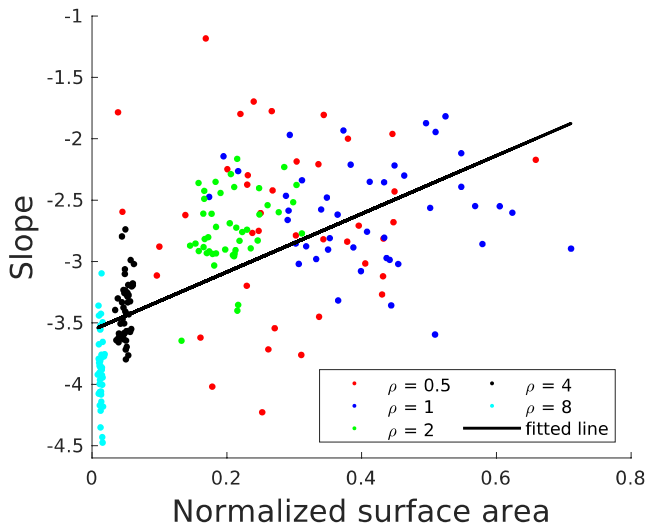


Figure 7. Scatter plot of the normalized surface area of dead-end fractures and tail slopes for $\rho = [0.5; 1; 2; 4; 8]$. A linear regression model is fitted to the data and the slope of the fitted line is 2.37, showing a positive correlation between dead-end fracture and late-time tailing.

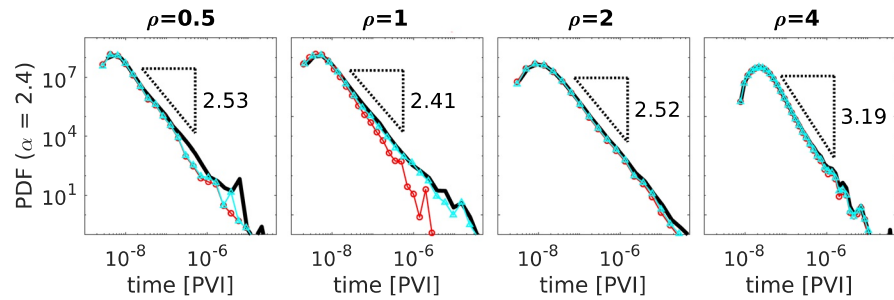


Figure 8. First passage time distributions (FPTDs) and backbone residence time distributions (RTDs) for all combinations of $\rho = [0.5; 1; 2; 4]$ when $\alpha = 2.4$. Red-colored lines with circle symbols: the RTDs obtained from using the proposed dead-end fracture definition. Cyan-colored lines with triangle symbols: the RTDs when the dead-end fractures are defined as the complement of the 2-core subgraph. The tail slopes are estimated by fitting a linear regression model to 95% and 99.99% quantiles.

area of dead-end fractures increases, providing further evidence that dead-end fractures exert an important control over late-time tailing. To quantify the correlation, we conducted a linear regression analysis using the data of tail slopes and normalized dead-end fracture area, as presented in Figure 7. The figure shows that the tail slope rises with increasing dead-end fracture's normalized area. The spread around the fitted line can be explained by the fact that the dead-end fractures are not the only mechanism for determining late-time tailing.

Sections 3.1 and 3.2 show that the relation between fracture density and solute transport (Figure 5) is similar to the relation between fracture density and dead-end fractures (Figure 6). Because network structure controls flow and transport behavior, it is reasonable to hypothesize that dead-end fractures play an important role in determining solute transport behavior. In the following section, we quantify the effects of dead-end fractures on solute transport.

3.3. Effects of Dead-End Fractures on Solute Transport

To investigate how dead-end fractures impact solute transport, we estimate the percentage of solute particles that travel through dead-end fractures, as shown in Figure 6b. Note that this analysis is feasible because particle transport simulation provides complete trajectories of solute particles. The particle percentage is maximum at the percolation threshold ($\rho = 1$), which is consistent with the solute tailing and dead-end fractures' surface area, $\bar{S}_d$. Therefore, dead-end fractures are at least a good indicator for solute spreading and late-time tailing.

To quantify the direct effects of dead-end fractures on solute tailing, we compute the residence times t_{dc} of the particles in the main body of the fracture network, excluding dead-end fractures. Here, the main body of the fracture network is the sub-network represented by G_{dc} . Thus, the main body residence time t_{dc} is related to the first passage time as $t_L = t_{dc} + t_d$, where t_d is the residence time in dead-end fractures. The main body residence times are nondimensionalized as $t'_{dc} = Q t_{dc} / V$, and the PDF of t'_{dc} , which we term backbone residence time distribution (RTD), is obtained by $p_{RTD}(t) = \langle \delta[t - t'_{dc}] \rangle$.

Figure 4 shows, along with FPTDs, the backbone residence time distributions (RTDs). By comparing the tails of FPTDs and backbone RTDs, we can directly quantify the contribution of dead-end fractures in inducing late-time tailing. The most significant discrepancy is observed at the percolation threshold (see the second column in Figure 4): the effects of particle residence time at dead-end fractures (the difference between black and the other lines) are maximized at the percolation threshold. At lower fracture densities, the solute residence times in the dead-end fractures contribute meaningfully to the late-time tailing. As the fracture density increases beyond the threshold, the backbone RTDs are almost identical to the FPTDs. As shown in Figure 6, the surface area of dead-end fractures at higher fracture density is smaller because the higher density increases network connectivity and limits the existence of dead-end fractures. Thus, residence times at dead-end fractures become less significant for high ρ , and this is why the backbone RTDs are almost identical to the FPTDs.

To demonstrate the effectiveness of the proposed dead-end fracture definition, we also present the backbone residence time distributions (RTDs) computed using the 2-core subgraph-based definition, as shown in Figure 8. While the backbone RTD obtained using the proposed dead-end fracture definition captures the reduced late-time tailing at $\rho = 1$, the backbone RTD computed using the 2-core subgraph-based dead-end fracture definition

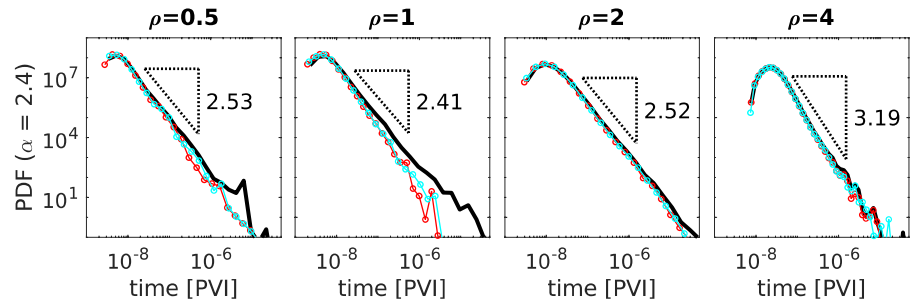


Figure 9. First passage time distributions from full DFNs (black-colored), backbone residence time distributions (red-colored), and first passage time distributions from the backbone networks (cyan-colored) for $\rho = [0.5; 1; 2; 4]$ at $\alpha = 2.4$. The tail slopes are estimated by fitting a linear regression model to 95% and 99.99% quantiles. The backbone RTDs and the FPTDs at backbone networks are aligned well.

is almost identical to the first-passage time distribution (FPTD). This result confirms that the newly proposed definition of dead-end fracture is better at capturing the trapping effects of dead-end fractures.

We now physically exclude dead-end fractures and run flow and transport simulations in the backbone networks. For this purpose, the dead-end fractures that are identified using the graph theory are removed, and flow and transport simulations are conducted on the backbone networks. Figure 9 shows the computed first passage time distributions (FPTDs) from the backbone networks as well as the FPTDs from full DFNs and backbone residence time distributions at $\alpha = 2.4$. The FPTDs obtained from the backbone network closely align with the backbone RTDs, providing further evidence that dead-end fractures play a significant role in inducing late-time tailing, especially at the percolation threshold.

The above results indicate that dead-end fractures directly contribute to late-time tailing at lower fracture densities, but their trapping role becomes less significant at higher fracture densities. The dead-end fractures' trapping effect is maximized at the percolation threshold, and the trends of dead-end fracture-related measures are highly correlated with the trends of solute spreading and tailing.

We now show the proposed dead-end fracture definition can also inform about the primary flow and transport paths because the fracture network that exclude dead-end fractures will form the backbone of fracture networks. Figure 10a shows the graph representation of the fracture network in Figure 3a. The vertices' size and the edges' width are proportional to the total number of particles traveled through them. Weighting the vertices and edges by particle counts helps easily identify the main flow and transport paths. Figure 10b shows the main transport path on the fracture network, where we show only the fractures corresponding to G_{dc} . The main body of the

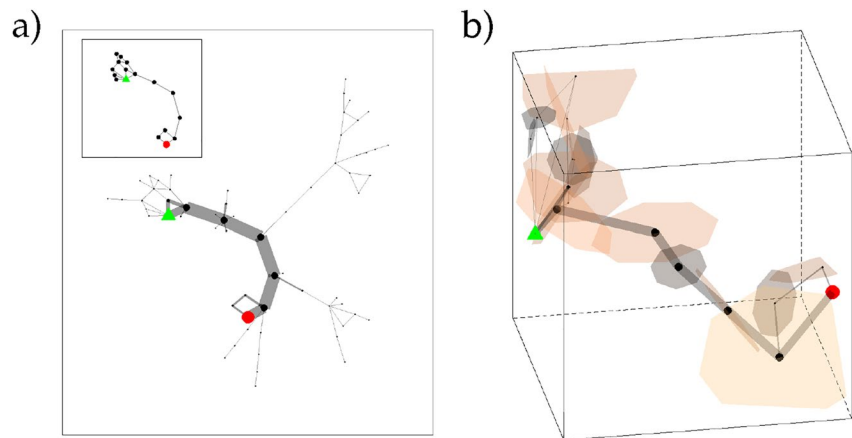


Figure 10. (a) Transport-weighted graph of the fracture network shown in Figure 2b. The inset figure shows the subgraph G_{dc} , excluding dead-end fractures. The subgraph G_{dc} aligns well with the highlighted transport path in the full graph. (b) The main body of the fracture network corresponding to G_{dc} .

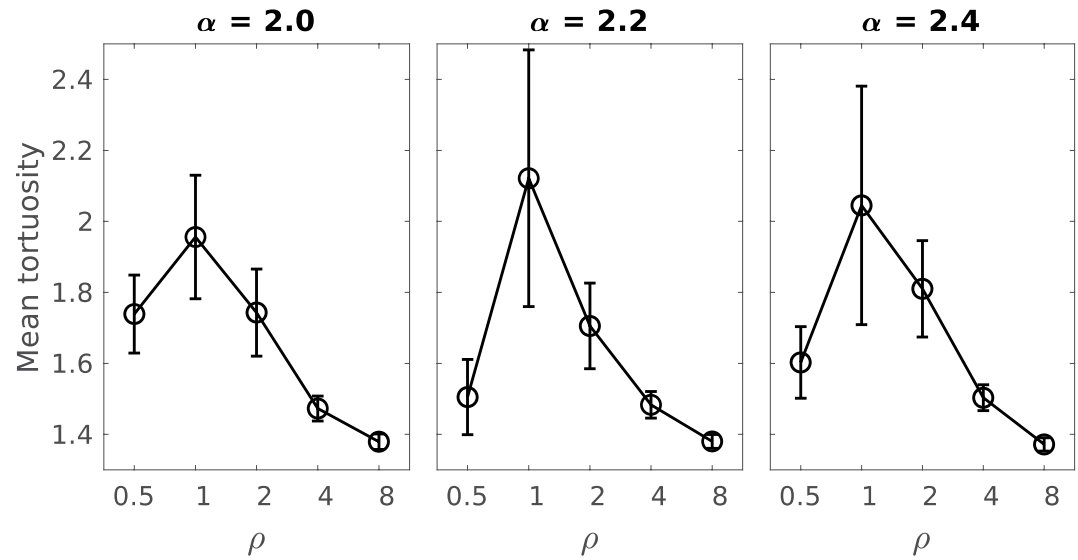


Figure 11. Mean and standard deviation of tortuosity for all combinations of α and ρ . The particle trajectories are computed by simulating particle transport in full fracture networks.

fracture network (G_{de}) is well aligned with the main transport path, demonstrating that the proposed definition of dead-end fracture well captures minor fractures and thus also the backbone network that excludes minor fractures.

It is worth noting that the dead-end fractures determined by the proposed topology-based method will be comparable to those that a flow-based method would identify. The inset of Figure 10a displays the subgraph G_{de} of fracture networks, excluding the dead-end fractures identified by the topology-based method. This subgraph aligns well with the major flow path computed through full flow simulation, as shown in Figure 10a. Since a flow-based method identifies low-flow fractures as dead-end fractures, the fractures that do not participate in the main flow path are considered dead-end fractures in the flow-based approach (Maillot et al., 2016).

Figure 10b shows that the main transport path is well-aligned with the backbone network (G_{de}). This indicates that the identification of dead-end fractures also helps capture the solute transport behavior dictated by the main transport path. The tortuosity of particle trajectories is a key measure that characterizes the main transport paths (Ghanbarian et al., 2013; Sherman et al., 2020). When the transport path becomes more tortuous, solute transport is likely to be delayed, and late-time tailing emerges. We compute the tortuosity of all particle trajectories as $\tau = L^{-1} \int_0^{t_L} |\mathbf{v}_t(t; \mathbf{a})| dt$, which is the average trajectory length divided by the domain size in the flow direction.

The error-bar plot in Figure 11 shows the mean and standard deviation of the tortuosity for all combinations of α and ρ . Both the mean and standard deviation of the tortuosity are maximized at the critical density of $\rho = 1$, indicating that the critical fracture density not only maximizes the tortuosity but also leads to a wide range of the tortuosity of transport paths. Further, the tortuosity trends are similar to the trends of dead-end fractures' surface area (Figure 6a), showing the strong correlation between tortuosity and dead-end fractures. This result suggests that dead-end fractures can serve as a good indicator of non-Fickian transport in fracture networks not only because they delay particle transport but also because they inform the tortuosity of main transport paths.

4. Summary and Conclusions

In this study, we elucidated how dead-end fractures influence non-Fickian transport in 3D fracture networks. To identify dead-end fractures, we first proposed a new topology-based definition of dead-end fractures as the fractures whose corresponding nodes in a graph G are the subset of the biconnected components whose removal does not disconnect the source and target vertices. The proposed definition comprehensively and efficiently identified dead-end fractures that do not comprise main transport paths. Because the proposed method requires only the topological information of fracture networks, it enables efficient identification of dead-end fractures compared to the previous flow-based methods that require flow information. Further, compared to the existing 2-core subgraph-based definition, the new definition is shown to better capture the trapping effects of dead-end

fractures. It is worth mentioning that alternative definitions of dead-end fractures can also be formulated, and the effectiveness of different definitions in capturing the influence of dead-end fractures on transport may vary.

By quantifying percolation thresholds, 3D DFNs are systematically generated over wide ranges of fracture density for different fracture length distributions. We then investigated how fracture density controls the existence of dead-end fractures in the generated 3D DFNs. Both the normalized dead-end fractures' surface area and the normalized number of dead-end fractures are maximized at the critical fracture density of the percolation threshold. Flow and particle transport simulations are conducted in the 3D DFNs, and the most significant solute tailing is also observed at the percolation threshold. At low fracture density, we demonstrate that dead-end fractures significantly increase the solute tailing by increasing the particle residence times within the dead-end fractures. We also found that the surface area of dead-end fractures is strongly correlated with the tortuosity of transport paths. Consequently, dead-end fractures are a good predictor for non-Fickian transport not only because they delay particle transport but also because they inform the tortuosity of main transport paths.

This study considers truncated power-law distributions for fracture length, a uniform distribution for fracture orientation, and a power-law correlation between fracture length and aperture. Various fracture network scenarios exist in the fields, and thus a wider range of fracture network scenarios needs to be investigated to further generalize the findings. In particular, aperture variability within a fracture is an important factor to consider in a future study. However, there is ample evidence in the literature from laboratory studies and simulations, both single fracture and network-scale, that there are circumstances where internal fracture aperture variability has little effect on upscaled flow and transport properties (de Dreuzy et al., 2012; Frampton et al., 2019; Makedonska et al., 2016). There are of course cases when that variability has a disproportionately large influence on the flow and associated transport (Huang et al., 2019). It is worth emphasizing that our study is applicable to the cases where such effects would be negligible. Additionally, incorporating aperture variability into high-fidelity simulations requires detailed knowledge of the rock formation, the fracture formation process, and any hydraulic and geochemical alteration, for example, dissolution and precipitation, to characterize and then synthetically produce a stochastic representation of the aperture field. Given that our study is meant to be semi-generic, this is infeasible.

In this study, diffusion was only effectively incorporated by assigning complete mixing at intersections (Kang, Dentz, et al., 2015; Park, de, et al., 2001), and we did not explicitly consider diffusive transport. In diffusive conditions, the degree of late-time tailing behavior would be enhanced because diffusion helps solute particles enter the low-velocity regions within dead-end fractures. Since this study only considers advection, we expect the key findings showing the importance of dead-end fractures should still be valid with diffusion. An important next step will be investigating the effects of dead-end fractures on solute transport in diffusive regimes. Despite the limitations, this study advances our fundamental understanding of the effects of dead-end fractures on non-Fickian transport in 3D DFNs. In particular, this study highlights the possibility of predicting non-Fickian transport directly from network structural features such as the density of dead-end fractures. Thus, this study contributes to the understanding of the connection between geostructural attributes and solute transport.

Acknowledgments

The authors acknowledge the support by the National Science Foundation under Grant EAR-2046015 and the support through the Korea Environmental Industry and Technology Institute (KEITI) funded by the Ministry of Environment (Grant RS-2023-00230833). We thank the Minnesota Supercomputing Institute (MSI) at the University of Minnesota for computational resources and support. J.D.H gratefully acknowledges support from the LANL LDRD program office Grant 20220019DR and Department of Energy (DOE) Basic Energy Sciences program (LANLE3W1) for support. Los Alamos National Laboratory is operated by Triad National Security, LLC, for the National Nuclear Security Administration of U.S. Department of Energy (Contract No. 89233218CNA000001) (LA-UR-22-28752).

Data Availability Statement

Computer codes for generating DFNs, running flow and transport simulation, and identifying dead-end fractures are online (Yoon et al., 2023).

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